Induction Heads and Model Collapse: How the Real-to-Synthetic Data Ratio Shapes In-Context Learning Circuits

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Abstract

Model collapse is the decline of models retrained on data generated by earlier models. To date, collapse has been measured through model outputs, which would understate the decline if the circuits behind those outputs degraded first. This study therefore traces the induction circuit behind in-context learning (ICL) in two-layer transformers retrained on data that the previous model partly or wholly relabels, or adds to the real data. A tuned model reproduces its own labels almost exactly, so that accuracy holds under every real-to-synthetic ratio. Once errors are forced through a raised sampling temperature, however, mechanistic changes precede accuracy loss: the logit margin halves and the circuit spreads while accuracy barely changes, and the margin predicts the next generation's error rate, so that errors compound. Retaining real data limits how far accuracy falls, and accumulation delays that fall, but neither stops the margin's erosion. Overall, real data protect a model's behaviour for longer than its circuit, so that collapse may show in the circuit first.

Keywords: model collapse; induction heads; mechanistic interpretability

Introduction

Recursively training large language models (LLMs) on data generated by other models causes *model collapse*, a loss of information about the true data distribution (Shumailov et al., 2024), which retaining real data appears to avoid (Gerstgrasser et al., 2024). Both findings, however, rest on model outputs, which would understate collapse if the underlying circuits degrade first.

In-context learning (ICL) lets models adapt to new tasks from prompt examples without parameter updates (Olsson et al., 2022; Singh et al., 2024). In the task of Singh et al. (2024), each sequence pairs symbols with arbitrary labels, so that the label of a query symbol can only be found by matching the query to the same symbol in the context and copying the label that follows (Figure S1). An induction circuit of known components performs this match and copy, so that damage can be traced to specific heads rather than inferred from accuracy. This study asks how the share of real data shapes this circuit.

Background

Induction heads A previous-token head in the first layer copies each symbol into the following label, and an induction head in the second layer matches the query against these copies and copies the matching label. Chen et al. (2026) trace the formation of these heads to specific training data, suggesting that the data mix can reshape the circuit.

Collapse and accumulation Collapse arises mainly from sampling error, which loses the tails of the distribution first and compounds across generations (Shumailov et al., 2024). Replacing real data makes test error grow each iteration, whereas adding synthetic data to the real data keeps the error bounded (Gerstgrasser et al., 2024), a contrast tested here in the circuit.

Methods

This study uses the setup of Singh et al. (2024), in which each sequence holds two symbol–label pairs and a query, with Omniglot exemplars (Lake et al., 2015) encoded by a frozen ResNet18 (He et al., 2016) and a set of five labels reassigned per sequence. A causal two-layer attention-only transformer with eight heads per layer trains on 50 classes for one million sequences, at learning rates of 10^{-4} (base) and 3×10^{-4} (extended) tuned on 1,473 validation classes, and is tested on 100 unseen classes. The base model predicts the query's label, and the extended model also predicts a next symbol and that symbol's label, so that three tokens per sequence become synthetic (Figure S1). The next symbol comes from the context in real data but from all 50 training classes in synthetic data.

Real-to-synthetic ratio Each run trains on a pool of $N = 20{,}000$ sequences for up to 20 generations, the first generation on real data only, with a fixed training length and three seeds per condition. The previous model then regenerates the targets of 0%, 50%, 90%, or 100% of the pool from their real prompts, where 90% matches the 10% of real data kept by Shumailov et al. (2024). Accumulation instead adds N synthetic sequences per generation (Gerstgrasser et al., 2024). Synthetic tokens are sampled from the softmax of the logits divided by a temperature τ, so that $\tau > 1$ flattens the distribution and makes wrong labels likelier. As the tuned model makes almost no errors at $\tau = 1$, errors are forced, or introduced deliberately, at $\tau = 3$ and 5.

Circuit analysis The model's margin, the logit of the correct label minus that of the other context label, measures how confidently the model answers. As the next generation's labels are sampled at τ, the mean margin D_g of generation g also predicts their error rate: taking D_g as the correct label's margin over each of the four wrong labels, sampling makes a share

$$\hat{\varepsilon}_{g+1} = \frac{4\,e^{-D_g/\tau}}{1+4\,e^{-D_g/\tau}} \tag{1}$$

of these labels wrong. Automatic Circuit DisCovery (ACDC) (Conmy et al., 2023) compares each sequence with a copy in which the two context symbols trade places, so that matching the query leads to the wrong label. ACDC keeps only edges whose corruption changes the output by more than $\theta = 0.001$. In the induction head, the match is the attention on the label after the matching symbol, and the copy is measured by the head's direct effect, the share of the model's margin that comes from this head's output alone.

Results

No errors, no collapse At generation 0, both models solve the task by ICL: the base model reaches a mean test accuracy of 0.98 and the extended model 0.96, with validation accuracies of 0.99 and 0.97 respectively. Without forced errors, the real-to-synthetic ratio barely matters, as at $\tau = 1$ only 0.01% of regenerated labels are wrong after 20 generations, so that every replacement level yields almost the same model (Table 1).

Table 1: Generation 19 of every 20-generation run, mean ± SD over three seeds; Real is the share of real data in the pool. †Margin below 1, where the circuit size is less reliable.

		Base model			Extended	
Condition	Real	Acc.	Margin	Edges	Margin	Edges
No forced errors ($\tau = 1$)						
Real only	100%	0.98 ±0.01	17.5 ±2.2	23 ±5	15.7 ±0.3	42 ±7
100% replaced	0%	0.98 ±0.00	12.5 ±0.3	35 ±7	12.8 ±0.9	92 ±36
Accumulate	5%	0.99 ±0.00	9.8 ±1.0	20 ±3	6.3 ±0.1	34 ±5
Forced errors ($\tau = 3$)						
50% replaced	50%	0.79 ±0.01	2.6 ±0.1	194 ±11	3.8 ±0.3	89 ±19
90% replaced	10%	0.33 ±0.01	0.4 ±0.1	207 ±3†	0.1 ±0.0	180 ±4†
100% replaced	0%	0.19 ±0.02	0.1 ±0.0	207 ±7†	−0.1 ±0.1	163 ±7†
Accumulate	5%	0.92 ±0.02	0.7 ±0.0	17 ±5†	0.6 ±0.0	11 ±2†

Confidence and circuit change before accuracy Once errors are forced at $\tau = 3$, the model's confidence and circuit change before the model starts answering wrongly. After one generation of full replacement, the margin has nearly halved, from 18.4 to 10.1, and the circuit has more than tripled, from 22 to 74 edges on average (17 and 55 in Figure 1), yet accuracy is still 0.95 or above. Only by generation 3, with the circuit spread across 211 edges, does accuracy fall, to 0.66 (Figure S3). The growth does not depend on ACDC's threshold, as the number of edges needed to recover 80% of the margin rises in the same way, from 33 to 120, while real-only retraining stays near 30.

The match holds, the copy weakens Inside the induction head, finding the match and copying the label fail at different times (Figure 2). Under full replacement, the head still attends to the matching label, keeping 88% of its generation 0 attention by generation 2, whereas its direct effect keeps only 41%, so that the head still finds the answer but copies the label far more weakly. The head's attention collapses only at generation 4, after accuracy has fallen. Under accumulation, attention stays at 0.89 or above for 15 generations while the direct effect falls by 86%, so that real data preserve the head's match but not its copy.

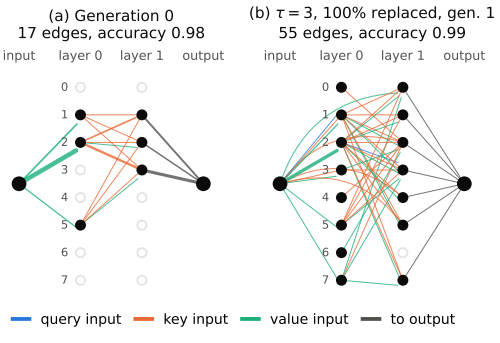


Figure 1: ACDC circuits of the base model (one seed; heads named A*layer.head*): previous-token head A0.2 feeds induction head A1.3 (a), and one generation at $\tau = 3$ triples the circuit (b).

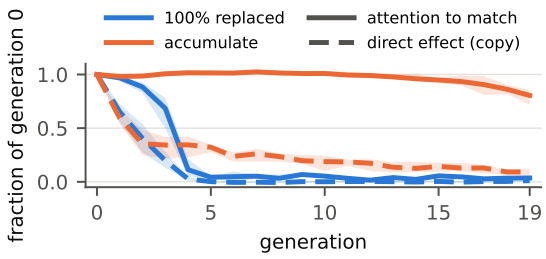


Figure 2: Induction-head match and direct effect at $\tau = 3$, mean of three seeds with the range shaded.

The margin sets the next generation's errors Equation 1 predicts the share of wrong labels in each new pool to within 2.1 percentage points on average across both models, temperatures, and conditions (correlation $r = 0.994$). A falling margin therefore writes more errors into the next pool, lowering the next margin, a loop that only real labels can slow.

The real share sets how far collapse goes Over 20 generations at $\tau = 3$, the more real data a pool keeps, the higher accuracy settles: 0.79 when half the pool stays real, 0.33 with a tenth, and 0.19 with none (Table 1, Figure S2). Yet even half leaves the circuit spread across 194 edges, so the real share bounds accuracy but not the circuit. Accumulation holds accuracy at 0.96 or above for 15 generations while the margin falls by 96%, and accuracy declines only once the margin is below 1. The extended model's query accuracy settles at 0.84, 0.34, and 0.20, with the same split in a second output. Under accumulation, its query accuracy stays at 0.93, but the raised temperature scatters its next symbols. At $\tau = 1$, its symbol head puts 13% of its probability on context symbols, against 98.6% under real-only retraining.

Discussion

The real-to-synthetic ratio shaped the induction circuit mainly once synthetic data carried errors. The share of real data set how far accuracy and the margin fell, as only real labels dilute the errors that a falling margin writes into each new pool, extending Shumailov et al. (2024) with the internal quantity that sets the sampling error. At every real share, however, the circuit changed before accuracy did: the margin fell, the circuit first spread, and within the induction head the copy weakened while the match held, suggesting that the copying operation of Singh et al. (2024) is the more fragile. Accumulation protected accuracy but not the margin or the copy, consistent with the bounded error of Gerstgrasser et al. (2024). These conclusions are bounded by forcing errors through temperature rather than a weaker model, and by a training length fixed while the accumulated pool grows 20-fold. A version that restricts the extended model's generated next symbols to the context, as in the real data, has been run, but its analysis is left for further work. Overall, real data protected a model's behaviour for longer than its confidence, so that the margin and the circuit could flag collapse before accuracy does.

Contribution Statements

We, Lily de Melo, Tawananyasha Kadango, and Gopolang Mohlakola, worked together fairly on this project.

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    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

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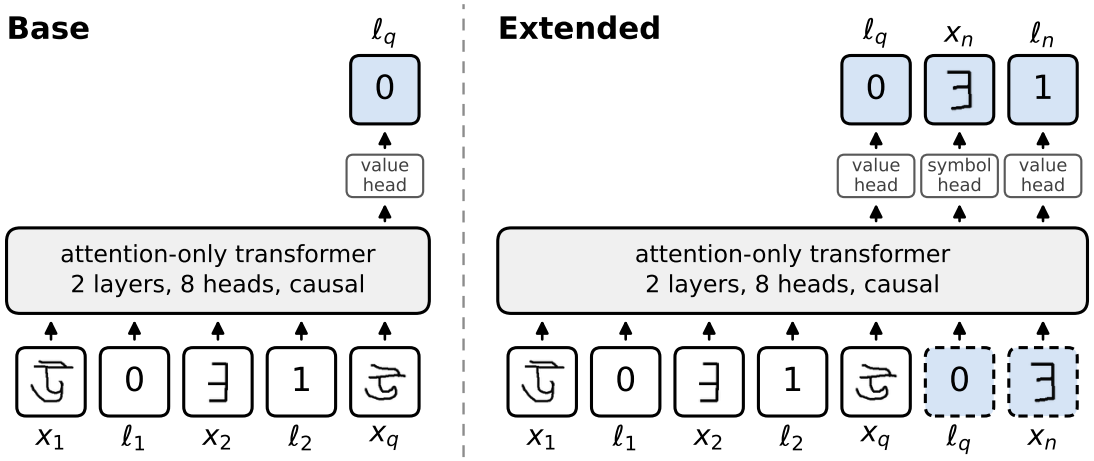


Figure S1: Base (left) and extended (right) models. The base model predicts the query's label from two symbol–label pairs and the query, and the extended model also predicts the next symbol and its label. Shaded boxes are model outputs; dashed inputs are true tokens during training and the model's own samples when it generates the next generation's data.

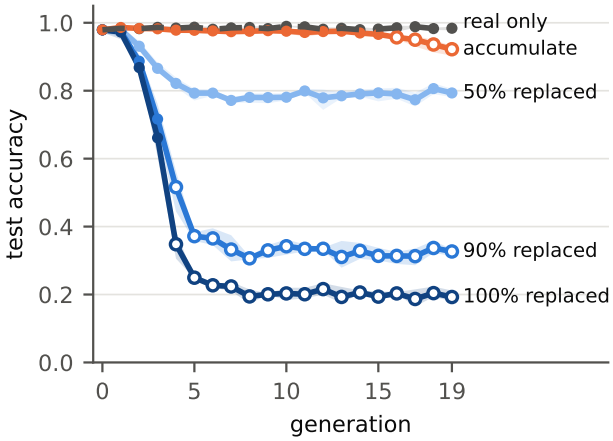


Figure S2: Test accuracy of the base model at $\tau = 3$, mean of three seeds with the range shaded. Each replacement level settles at an accuracy set by its real share, while accumulation stays with the real-only control for about 15 generations before declining. Hollow markers show a mean margin below 1.

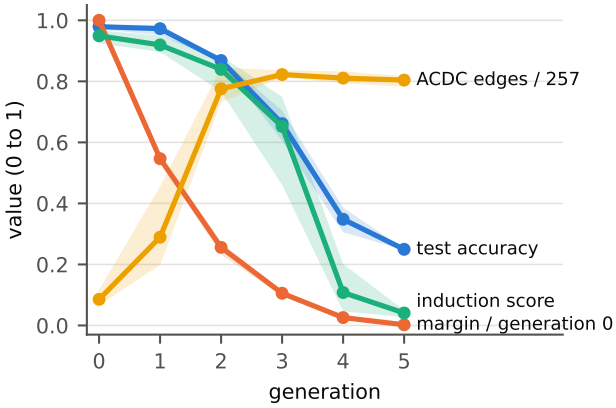


Figure S3: Order of failure of the base model under full replacement at $\tau = 3$, mean of three seeds with the range shaded. The margin halves and the circuit more than triples by generation 1, whereas accuracy and the induction score fall mainly from generation 3.

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I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools ☐

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1.	Idea Generation (research problem/design, hypothesis)	☐
2.	Sourcing Related Work (summarising, identifying sources)	☐
3.	Methods and Experiment Design (experiment setup, model tuning)	X
4.	Data Analysis (presentation, coding, interpretation)	X
5.	Theoretical Development (theorem proving, conceptual analysis)	☐
6.	Code Development (generating algorithms, writing scripts)	X
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